

Algebra 1
Unit 3
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

Use the equation $f(x) = \frac{5}{3}x - 1$ for questions 1 and 2.

1. What is the form of this linear equation?

This equation is in slope-intercept form.

2. Rewrite this equation in standard form.

$$y = \frac{5}{3}x - 1$$
$$\frac{-5}{3}x \quad \frac{-5}{3}x$$

$\frac{-5}{3}x + y = 1$ or any equivalent equation in standard form.

Use the equation $y + 1 = -\frac{1}{3}(x - 3)$ for questions 3 and 4.

3. What is the form of this linear equation?

This equation is in point-slope form.

4. Rewrite this equation in standard form.

$$y + 1 = \frac{-1}{3}(x - 3)$$
$$y + 1 = \frac{-1}{3}x + 1$$
$$+\frac{1}{3}x \quad +\frac{1}{3}x$$

$$\frac{1}{3}x + y + 1 = 1$$

$\frac{1}{3}x + y = 0$ or any equivalent equation in standard form.

Use the equation $6x + 2y = -8$ for questions 5 and 6.

5. What is the form of this linear equation?

This equation is in standard form.

6. Rewrite this equation in slope-intercept form.

$$6x + 2y = -8$$
$$-6x \quad -6x$$

$$\frac{2y}{2} = \frac{-6x}{2} - \frac{8}{2}$$

$$y = -3x - 4$$

Use the equation $y + 3 = -2(x + 7)$ for questions 7 through 10.

7. What is the slope of the line?

The slope is -2 $y - y_1 = m(x - x_1)$

8. Rewrite this equation in slope-intercept form.

$$\begin{array}{l} y + 3 = -2(x + 7) \\ y + 3 = -2x - 14 \\ \quad -3 \qquad \quad -3 \end{array}$$

$y = -2x - 17$

$y = mx + b$

↓ ↓
m b

9. Identify the y -intercept of the line.

The y -intercept is $(0, -17)$

10. Rewrite the equation in standard form.

$y = -2x - 17$
 $+2x \qquad +2x$

$2x + y = -17$ or any equivalent equation in standard form.